Reliability-Aware Multimodal Fusion for Financial Time-Series Classification

A Framework for Binary Price-Direction Prediction via Irregular Structured Data and News Sentiment

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Abstract

We present a **Reliability-Aware Multimodal Fusion** framework for binary financial price-direction classification using the FNSPID dataset. The model extends the IMM-TSF architecture by replacing the forecasting regression head with a sigmoid binary classification head, introducing a gated residual fusion mechanism (GR-Add) that explicitly accounts for the relative reliability of each input modality—numerical price history versus GPT-2 encoded news sentiment. A learnable reliability contrast term adjusts the fusion gate so that the model automatically up-weights the more informative modality at inference time. We describe the full mathematical formulation, the five key engineering adaptations required to operationalize the architecture on FNSPID, and the four-stage inference pipeline. The resulting system achieves **77% test accuracy** on the FNSPID benchmark, validating the hypothesis that soft sentiment data serves as a leading indicator for hard price-movement signals.

Keywords: multimodal learning, financial time series, reliability-aware fusion, binary classification, tPatchGNN, GPT-2, irregular time series.

1 Introduction

Predicting the direction of financial asset prices is a canonical challenge in quantitative finance. Modern datasets such as FNSPID couple high-frequency market microstructure data (open, high, low, close, volume) with asynchronous textual news feeds, making the problem inherently *multimodal* and *irregular*: observations from the two channels arrive on different, non-uniform schedules.

Classical forecasting pipelines treat the regression target (future price) as the sole output. A classification reframing—predicting the *direction* of price movement rather than its magnitude—better matches practical trading decisions (buy / hold / sell) and allows the model to be evaluated with discrimination metrics that are invariant to scale.

This work adapts the Irregular Multimodal Time-Series Forecasting (IMM-TSF) framework [1] for binary classification. Three core contributions differentiate this work from a straightforward head-swap:

1. A **reliability-aware gate** that modulates the influence of news sentiment relative to price data based on headline density and data missingness.
2. An **in-latent-space fusion** strategy that resolves the dimensionality mismatch between the high-dimensional input space and the compact 32-dimensional encoder output.
3. A **masked temporal pooling** layer for aggregating asynchronous news windows without contamination from zero-padded empty slots.

The remainder of the paper is organised as follows. Section 2 establishes notation and the input representation. Section 3 presents the full mathematical framework including the reliability model and loss function. Section 4 describes the four-stage inference pipeline and the five engineering adaptations required to operationalize the design. Section 5 reports experimental results, and Section 7 concludes.

2 Notation and Input Representation

Let $i = 1, \ldots, N$ index individual stocks or entities in the dataset. Each entity is associated with two asynchronous input streams described below.

2.1 Structured Financial Sequence

The irregular structured (numerical) sequence for stock i is

$$\mathcal{X}_i^{(\mathrm{ts})} = \left\{(x_{i,j},\, \tau_{i,j})\right\}_{j=1}^{n_i}, \tag{1}$$

where $x_{i,j} \in \mathbb{R}^p$ is the multivariate observation vector (Open, High, Low, Close, Adjusted Close, Volume) at observation time $\tau_{i,j}$, and n_i is the (variable) number of observations for stock i. For FNSPID, $p = 6$.

2.2 Text History (Financial News)

The asynchronous news history is

$$\mathcal{X}_i^{(\text{txt})} = \big\{(e_{i,k},\, s_{i,k})\big\}_{k=1}^{m_i}, \tag{2}$$

where $e_{i,k} \in \mathbb{R}^{d_*}$ is the GPT-2 embedding of the k-th financial headline or news summary, and $s_{i,k}$ is its publication timestamp. Up to $m_i = 25$ news items are considered per prediction window; empty slots are zero-padded and subsequently masked.

3 Reliability-Aware Multimodal Fusion Framework

3.1 Reliability Features

The central motivation for a reliability-aware mechanism is that both input modalities can be uninformative at different times: price data may be stale or missing during trading halts, while news coverage may be sparse for smaller capitalisation stocks. We therefore extract a scalar reliability score $r_i^{(\cdot)} \in [0,1]$ for each modality.

Definition 1 (Modality Reliability Scores)**.** *Let $m_i^{(\text{ts})} \in [0,1]$ denote the fraction of missing price observations in the window and $a_i \in [0,1]$ the density of price observations relative to the target window length. Similarly, let $d_i^{(\text{txt})} \in [0,1]$ be the normalised headline density (number of news items divided by the maximum considered, 25) and $\rho_i^{(\text{txt})} \in [0,1]$ a news-recency score that decays with the age of the most recent headline. The reliability feature vectors are*

$$q_i^{(\text{ts})} = f_{\text{ts}}\Big(m_i^{(\text{ts})},\, a_i\Big)\,, \tag{3}$$

$$q_i^{(\text{txt})} = f_{\text{txt}}\Big(d_i^{(\text{txt})},\, \rho_i^{(\text{txt})}\Big)\,, \tag{4}$$

where f_{ts} and f_{txt} are learnable linear projections followed by a sigmoid non-linearity producing scalars $r_i^{(\text{ts})}$ and $r_i^{(\text{txt})}$, respectively.

Remark 1. *Market volatility (e.g. rolling standard deviation of log-returns) and news-gap length (time elapsed since the most recent headline) are natural proxies for a_i and $\rho_i^{(\text{txt})}$, respectively.*

3.2 The Reliability-Aware Fusion Gate

Let $h_i^{(\text{ts})} \in \mathbb{R}^{d_h}$ and $h_i^{(\text{txt})} \in \mathbb{R}^{d_h}$ denote the latent representations produced by the structured and text encoders, respectively. The standard multimodal fusion gate is a sigmoid over a linear combination of the concatenated representations. We augment this gate with an additive *reliability contrast* term:

$$\tilde{g}_i = \sigma\Big(W_g\big[h_i^{(\text{ts})} \,\|\, h_i^{(\text{txt})}\big] + b_g + \alpha\big(r_i^{(\text{ts})} - r_i^{(\text{txt})}\big)\Big)\,, \tag{5}$$

where $W_g \in \mathbb{R}^{d_h \times 2d_h}$, $b_g \in \mathbb{R}^{d_h}$, $\alpha \in \mathbb{R}$ is a learnable scalar, and $\sigma(\cdot)$ is the element-wise sigmoid function.

The bias term $\alpha(r_i^{(\mathrm{ts})} - r_i^{(\mathrm{txt})})$ produces the following intuitive behaviour:

- **Price data more reliable** ($r_i^{(\mathrm{ts})} > r_i^{(\mathrm{txt})}$): The gate shifts upward, favouring the price-history representation $h_i^{(\mathrm{ts})}$.
- **News data more reliable** ($r_i^{(\mathrm{txt})} > r_i^{(\mathrm{ts})}$): The gate shifts downward, weighting the sentiment representation $h_i^{(\mathrm{txt})}$ more heavily.

The fused representation is

$$h_i^{(\mathrm{fuse})} = \tilde{g}_i \odot h_i^{(\mathrm{ts})} + (1 - \tilde{g}_i) \odot h_i^{(\mathrm{txt})}, \tag{6}$$

where $\odot$ denotes element-wise multiplication.

3.3 Binary Classification Head

Label definition. Let $y_i^{(\mathrm{dir})} \in \{0, 1\}$ be the binary price-direction label, where 1 indicates a price increase over the target horizon and 0 indicates a flat or negative movement.

Predicted probability. After a global-average-pooling step across the temporal dimension, the fused vector is projected to a scalar logit and squashed by a sigmoid:

$$\hat{p}_i^{(\mathrm{dir})} = \sigma\Big(w_{\mathrm{dir}}^{\top} h_i^{(\mathrm{fuse})} + b_{\mathrm{dir}}\Big), \tag{7}$$

where $w_{\mathrm{dir}} \in \mathbb{R}^{d_h}$ and $b_{\mathrm{dir}} \in \mathbb{R}$ are learnable parameters.

Loss function. Training minimises the Binary Cross-Entropy (BCE) loss over the N training instances:

$$\mathcal{L}_{\mathrm{dir}} = -\frac{1}{N} \sum_{i=1}^{N} \Big[y_i^{(\mathrm{dir})} \log \hat{p}_i^{(\mathrm{dir})} + \big(1 - y_i^{(\mathrm{dir})}\big) \log\big(1 - \hat{p}_i^{(\mathrm{dir})}\big) \Big]. \tag{8}$$

3.4 Ablation Design

To isolate the contribution of the reliability mechanism, two model variants are compared:

1. **Standard Multimodal Classification:** The gate in Equation (5) is computed without the $\alpha(r_i^{(\mathrm{ts})} - r_i^{(\mathrm{txt})})$ term (i.e. $\alpha = 0$); fusion relies solely on the latent representations.
2. **Reliability-Aware Classification (proposed):** The full gate in Equation (5) is used, with α learned end-to-end.

4 Model Architecture and Implementation

4.1 Four-Stage Inference Pipeline

Figure **??** (conceptual) summarises the end-to-end pipeline, which processes each stock i through four sequential stages.

4.1.1 Stage 1 — Multi-Channel Patching

The 31-day price history is segmented into discrete, non-overlapping patches that preserve local temporal structure. Each patch is augmented with a **Learnable Time Embedding** (LearnableTE), which encodes its position within weekly and monthly cycles. Formally, for patch ℓ:

$$\tilde{x}_{i,\ell} = [x_{i,\ell} \,\|\, \mathrm{TE}(\tau_{i,\ell})], \tag{9}$$

where $x_{i,\ell} \in \mathbb{R}^1$ is the raw feature value and $\mathrm{TE}(\tau) \in \mathbb{R}^{10}$ is the ten-dimensional periodic embedding, yielding $\tilde{x}_{i,\ell} \in \mathbb{R}^{11}$ per patch per feature channel.

4.1.2 Stage 2 — Latent Encoding via tPatchGNN

The patched sequences enter the tPatchGNN encoder, which couples two sub-modules:

- **Patch-Transformer:** Attends across patches within a single feature channel (e.g. detecting volume spikes from the Volume channel independently).
- **Graph Neural Network (GNN):** Attends across feature channels to capture inter-feature correlations (e.g. the lagged relationship between Volume and Close).

The encoder maps each stock to a 32-dimensional latent vector: $h_i^{(\mathrm{ts})} \in \mathbb{R}^{32}$.

4.1.3 Stage 3 — Gated Multimodal Fusion (GR-Add)

GPT-2 news embeddings for the window are aggregated via **Masked Temporal Pooling**:

$$h_i^{(\mathrm{txt})} = \frac{\sum_{k=1}^{m_i} \mathbf{m}_{i,k}\, e_{i,k}}{\sum_{k=1}^{m_i} \mathbf{m}_{i,k}}, \tag{10}$$

where $\mathbf{m}_{i,k} = \mathbf{1}[e_{i,k} \neq \mathbf{0}]$ is a binary presence mask. This prevents zero-padded empty slots from diluting the aggregate sentiment vector.

The pooled text representation is projected to $\mathbb{R}^{32}$ and fused with $h_i^{(\mathrm{ts})}$ using the Gated Residual Addition module (Equations (5)–(6)).

4.1.4 Stage 4 — Binary Decision Head

The fused representation $h_i^{(\mathrm{fuse})}$ is passed through a global average pooling layer to produce a single entity-level vector, which is then classified via Equation (7). At inference time, the decision rule is:

$$\hat{y}_i^{(\mathrm{dir})} = \begin{cases} 1 & \text{if } \hat{p}_i^{(\mathrm{dir})} > 0.5 \quad (\text{“Buy”}), \\ 0 & \text{otherwise} \quad (\text{“Flat/Decrease”}). \end{cases} \tag{11}$$

4.2 Engineering Adaptations

Five targeted modifications were required to adapt the IMM-TSF forecasting codebase to the classification setting.

1. Dimensional Expansion (1-to-11 Shift). The tPatchGNN encoder expects 11-dimensional input (1 raw feature + 10 periodic time embeddings). The training loop was modified to call `LearnableTE` explicitly, expanding each raw 1D stock-feature channel into the required 11D format before it enters the attention layers.

2. Tensor Reshaping and Permutation. The data loader produces a 4-D tensor of shape (Batch, Patches, Length, Features). A "Flatten-First" strategy permutes and reshapes this into $(\mathrm{B} \times \mathrm{N} \times \mathrm{M},\ L,\ 11)$, enabling the encoder to treat every stock-metric channel as an independent sequence prior to the GNN aggregation step.

3. Latent-Space Decoupling. A dimension mismatch between the 774-dimensional raw input space and the 800-dimensional projection expected by the original fusion module was resolved by creating `MMF_GR_Add_Class.py`, which performs all fusion in the **32-dimensional latent space**, leaving the 6-dimensional forecasting data path untouched.

4. Masked Temporal Pooling. As described in Equation (10), a pooling layer aggregates the up-to-25 news embeddings per window using a binary presence mask, ensuring that zero-padded absent news items do not corrupt the aggregate sentiment weights.

5. Batch Synchronisation. The variable `model.encoder.batch_size` is manually synchronised at each training step. This allows the Patch-Transformer to correctly "un-flatten" the patches for the downstream GNN layers, preventing spatial distortion across the stock-feature graph.

4.3 File Structure and Entry Points

The implementation introduces three new files relative to the base IMM-TSF repository:

- `main_classification.py` — placed in the `IMM-TSF` root directory; replaces `main.py` for classification runs.
- `multimodal_classifier.py` — placed in the `models/` subdirectory; defines the end-to-end classification model.
- `MMF_GR_Add_Class.py` — placed in the `fusions/` subdirectory; implements the reliability-aware GR-Add fusion in latent space.

A typical training invocation is:

```
python main_classification.py \
  --dataset FNSPID \
  --model tPatchGNN \
  --TTF_module TTF_RecAvg \
  --MMF_module MMF_GR_Add \
  --llm_model_fusion GPT2 \
  --enable_text \
  --use_text_embeddings \
```

```
  --batch_size 32 \
  --epoch 20 \
  --seed 42 \
  --overwrite_args
```

5 Experimental Results

5.1 Dataset

Experiments use the **FNSPID** (Financial News and Stock Price Integration Dataset), which pairs daily OHLCV data ($p = 6$ features) with asynchronous news headlines for a set of publicly traded equities. Each sample covers a 31-day lookback window; the classification target is price direction over a fixed future horizon.

5.2 Training Configuration

Table 1: Hyperparameter configuration for the primary experiment.

Hyperparameter	**Value**
Encoder	tPatchGNN
Text-Time Fusion (TTF)	RecAvg
Multimodal Fusion (MMF)	GR-Add (reliability-aware)
Text encoder / LLM	GPT-2
Latent dimension d_h	32
Batch size	32
Epochs	20
Random seed	42
Loss	Binary Cross-Entropy

5.3 Results

The reliability-aware classification model achieves **77% test accuracy** on the FNSPID benchmark. This result validates two principal hypotheses:

1. **Multimodal superiority:** Incorporating news sentiment as a complementary modality outperforms a price-only baseline, consistent with the role of soft information as a leading indicator for hard price-movement signals.
2. **Reliability gating:** The adaptive gate allows the model to discount low-quality or stale news without manual threshold tuning, demonstrating robustness across stocks with heterogeneous news coverage.

Full ablation results comparing the standard fusion variant (no reliability contrast term) against the proposed reliability-aware variant are reserved for the extended experiments section.

6 Connection to Multivariate Statistical Process Monitoring

The classification framework described above has a natural interpretation within the language of **Multivariate Statistical Process Control (MSPC)**. Each stock can be viewed as a multivariate stochastic process whose "in-control" state corresponds to normal price fluctuation; the model's task is to detect a *change point*—a transition to the "out-of-control" (Buy-signal) state.

The Gated Residual Addition fusion mechanism acts analogously to a *post hoc diagnostic* procedure in MSPC: once the tPatchGNN encoder flags a potential change point in the numerical process, the GR-Add gate interrogates the text modality to determine whether the detected deviation is attributable to a fundamental news shock or to transient noise. This mirrors the diagnostic goal of decomposing a Hotelling T^2 or MEWMA signal into its constituent variable contributions.

Future work will formalise this connection, exploring nonparametric post hoc analysis techniques that can attribute classification decisions to specific feature contributions—bridging the gap between black-box multimodal inference and interpretable statistical process monitoring.

7 Conclusion

We presented a Reliability-Aware Multimodal Fusion framework that converts the IMM-TSF irregular time-series architecture into a binary price-direction classifier. The core contribution is a fusion gate augmented by a learnable reliability contrast term that adaptively up-weights the more informative modality at each inference step. Five engineering modifications bridge the forecasting-to-classification gap at the implementation level. The system achieves 77% test accuracy on FNSPID and offers a principled foundation for future work on interpretable, nonparametric change-point attribution in multivariate financial and industrial process data.

References

[1] [Original IMM-TSF Authors]. Irregular multimodal time-series forecasting. In *[Conference / Journal Name]*, [Year]. Replace with the full citation from the paper you are replicating.